

HVAC Building Energy Management Multi-Objective Deep Reinforcement Learning

Project Defense Presentation: Phases 0–3 and Current Phase 3 Continuation

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Agenda

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Problem Statement

- HVAC control is inherently multi-objective: comfort, energy, and safety must be balanced at the same time.
- Purely model-based methods are often accurate but slow or difficult to deploy online.
- Purely model-free RL is flexible, but BOPTEST training is expensive and sample-inefficient.
- The central question of this project is therefore:
- Can we build a fast and physically informed RL pipeline that is accurate enough for control, scalable enough for experimentation, and structured enough for safe deployment?

- Phase 0: establish PPO-based energy-management baseline and identify reward-scaling and observation-space issues.
- Phase 1: build a physics-informed RC Neural ODE surrogate to replace slow BOPTEST rollouts.
- Phase 2: calibrate the thermal model through an inverse problem to improve physical consistency.
- Phase 3: test increasingly structured control architectures, culminating in HDRL, real weather, emergency logic, and now the current direct- T_{Sup} continuation.

Phase 0: Initial MORL Baseline

- Initial PPO-based baseline established the first Pareto trade-off between comfort and energy.
- Two critical issues were discovered:
 - scale dominance in the scalarized reward,
 - observation-space mismatch between code and environment.
- After fixing these, the early baseline achieved a reported 28% energy reduction and produced the first usable Pareto front.
- Phase 0 was important because it revealed that most early instability was not algorithmic, but representational.

Phase 1: Why Neural ODE?

- BOPTEST is accurate but slow; large RL sweeps directly on the simulator are impractical.
- The solution was to learn a surrogate with a physically interpretable thermal update.
- The implemented model is an RC Neural ODE with two heads:
 - a thermal head predicting ΔT ,
 - a power head predicting total HVAC power.
- This preserves the physical meaning of the state transition while remaining fast enough for massive batched rollouts.

Neural ODE Formulation

The current surrogate implementation in `surrogate/rc_node_v2.py` uses the feature vector

$$\mathbf{x}_t = [\tilde{T}_{\text{zone}}, \tilde{T}_{\text{amb}}, \sin h, \cos h, \sin d, \cos d, a_0, a_1].$$

It predicts:

$$\Delta T_t = f_{\theta}(\mathbf{x}_t), \quad P_t = g_{\phi}(\mathbf{x}_t),$$

and updates temperature as

$$T_{t+1} = \text{clip}(T_t + \Delta T_t).$$

Key design choices:

- residual connection in the heat network,
- cyclic encodings for hour and day,
- explicit power positivity via Softplus,
- multi-step training loss with physical regularization.

Phase 1 Results

- Reported speed:
 - BOPTEST: 37.6 steps/s,
 - surrogate batch rollout: 1,215,752 steps/s,
 - speedup: $32,313\times$.
- The Phase 3 surrogate-v2 report further improved one-step RMSE from 1.128 to 0.163 °C with $R^2 = 0.991$.
- This surrogate became the computational foundation for the later PPO, HDRL, and safety-filter pipelines.

Phase 2: Inverse-Problem Calibration

- The inverse problem stage addressed parameter mismatch between the nominal thermal model and the observed building response.
- Two key issues were solved:
 - latency and ill-posed preprocessing,
 - weak linear calibration.
- Reported improvement:
 - RMSE reduced from 9.12 to 1.91 °C,
 - 79% error reduction,
 - thermal capacitance error reduced to 7.9%.
- This phase made the surrogate more physically trustworthy, which matters later for safety and for control transfer.

Phase 3 in the Final Report

According to Mor1ProjectFinalv3.pdf, Phase 3 consisted of ten systematic experiments.

Main reported findings:

- HDRL specialization improved over flat PPO.
- Real weather integration removed a major simulator-to-controller mismatch.
- Emergency heating solved many extreme cold failures.
- Reward shaping alone amplified the gap instead of closing it.
- BOPTEST fine-tuning did not solve the remaining issue, indicating a structural mismatch rather than a pure distribution shift.

Best Reported Legacy Phase 3 Configuration

Method	m_s	Viol. %	Main Finding
HDRL + RW + Emerg + 5F + [18,30]	0.704	46.8	Final config in report
HDRL + RW + Emerg (4F, [21,25])	0.697	45.8	Best m_s in report
PPO baseline (synthetic)	0.798	51.9	Strong baseline
GRU (Gao)	0.893	57.4	Forecast gap

Legacy final controller:

- 5-feature observation,
- hierarchical winter/summer agents,
- emergency rule for severe cold,
- action mapped to a setpoint-like control in the range [18,30] °C.

Why Phase 3 Had to Continue

The final report itself documented that the remaining performance gap was structural.

The root-cause analysis highlighted three facts:

- weather mismatch was solved,
- extreme cold events were partly solved,
- the remaining gap came from action-response mismatch.

This directly motivated the current continuation:

- move from indirect thermostat-style control to direct supply-air control,
- unify surrogate training, PPO training, and BOPTEST evaluation under the same action semantics,
- re-establish strong baselines before re-launching MORL.

Current Redesign: Direct Supply-Air Control

The current continuation changes the control interface fundamentally.

Previous logic:

- indirect thermostat/setpoint control,
- fixed or semi-fixed supply-air behavior,
- higher risk of mismatch between surrogate and BOPTEST.

New logic:

- direct control of T_{Sup} in the range $[18,35]$ °C,
- direct fan command,
- wide heating/cooling setpoints in BOPTEST to keep the RL controller in charge,
- identical action semantics in surrogate and BOPTEST backends.

Current End-to-End Pipeline

- Data collection: `data/collect_tsupply_data.py`
 - $4 \text{ seasons} \times 4 \text{ policies} \times 3200 \text{ steps} = 51,200 \text{ samples}$.
- Surrogate training: `surrogate/train_surrogate_v2.py`
 - multi-step loss, physics regularization, and power prediction.
- Thermostatic comfort-first PPO: `training/train_thermostatic.py`
- Seasonal HDRL experts: `training/train_hdrl.py`
- Standard PI reference: `evaluation/standard_controller_baseline.py`
- Ongoing MORL and safety layer:
 - `main.py`,
 - `evaluation/eval_safe_morl.py`,
 - `layers/safety/action_filter.py`.

Standard PI Controller as the Energy Reference

The built-in BOPTEST controller is now evaluated explicitly without any override signals.

Two comfort views are reported:

- fixed-target metrics against 22 °C for direct comparison with PPO-based baselines,
- schedule-aware metrics using the testcase schedule from `internal_setpoints_occupancy.csv`.

Mean results over 12 benchmark windows:

- $\text{RMSE}_{22} = 3.41 \text{ }^{\circ}\text{C}$,
- fixed-band violation = 43.8%,
- schedule-aware violation = 27.0%,
- schedule-aware $m_s = 0.503$,
- total energy = 1230.2 kWh = 25.63 kWh/m².

Why the PI Baseline Matters

- The standard controller is not the best comfort controller for a strict 22 °C target.
- However, it is the most energy-efficient reference available in the current benchmark.
- Therefore, it should be interpreted as:
 - the low-energy reference,
 - not the main apples-to-apples comfort baseline.
- This gives the project a more honest evaluation frame:
 - PI baseline = energy reference,
 - thermostatic PPO = comfort reference,
 - HDRL/MORL = trade-off controllers.

Current Thermostatic PPO Baseline

The current comfort-first baseline is implemented in `training/train_thermostatic.py`.

Main design choices:

- training on the direct- T_{Sup} surrogate `rc_node_v3_tsupply.pt`,
- 17-dimensional observation,
- direct ambient-aware reward centered on the exact target 22 °C,
- weak energy penalty only near the target,
- 10 million PPO timesteps with 32 parallel environments.

This controller is intentionally optimized for comfort, not for energy minimization.

Thermostatic PPO: Current Validated Results

Metric	Value	Interpretation
Mean RMSE	0.97 °C	strong annual comfort
Mean MAE	0.81 °C	stable tracking
Within ± 1 °C	65.9%	close to target most of the time
Within ± 0.5 °C	35.1%	reasonably tight control
Total energy	3037.8 kWh	comfort-first cost
Normalized energy	63.29 kWh/m ²	much higher than PI

Seasonal RMSE:

- winter: 0.92 °C,
- spring: 0.89 °C,
- summer: 1.20 °C,
- autumn: 0.87 °C.

Interpretation of the Thermostatic Result

- This is the strongest comfort result currently validated in the repository.
- The previous winter failure mode of the thermostat-style pipeline is largely removed.
- The remaining weakness has shifted toward cooling-season performance, not heating collapse.
- This controller now serves as the main comfort benchmark for all later trade-off methods.

The current HDRL implementation in `training/train_hdrl.py` uses:

- two PPO experts trained on the direct- T_{Sup} surrogate,
- a compact 5-feature observation:

$$[T_{\text{zone}}, CO_2, P_{\text{total}}, T_{\text{Sup,prev}}, T_{\text{amb}}],$$

- winter and summer specialization,
- real weather-based domain randomization,
- emergency heating in evaluation when $T_{\text{amb}} < 5\text{ }^{\circ}\text{C}$ and $T_{\text{zone}} < 20\text{ }^{\circ}\text{C}$.

The meta-controller currently switches by a simple ambient threshold:

$$T_{\text{amb}} < 12^{\circ}\text{C} \Rightarrow \pi_{\text{winter}}, \quad \text{otherwise } \pi_{\text{summer}}.$$

HDRL: Current Validated Results

Metric	Value	Interpretation
Mean violation rate	12.7%	reasonable but not comfort-optimal
Mean m_s	0.307	better than old reported HDRL
Total energy	3070.2 kWh	slightly above thermostatic baseline
Normalized energy	63.96 kWh/m ²	far above PI baseline
Winter mean violation	20.6%	main weakness
Summer mean violation	16.1%	overcooling still visible

Summer-only energy:

- thermostatic PPO: 422.7 kWh,
- HDRL: 404.6 kWh,
- summer saving of HDRL vs thermostatic: 4.3%.

Controller Comparison

Controller	Comfort View	Energy	Role
Standard PI	$\text{RMSE}_{22} = 3.41\text{ }^{\circ}\text{C}$	1230.2 kWh	energy reference
Thermostatic PPO	$\text{RMSE} = 0.97\text{ }^{\circ}\text{C}$	3037.8 kWh	comfort reference
HDRL	$m_s = 0.307$, viol = 12.7%	3070.2 kWh	trade-off prototype

Relative energy versus PI baseline:

- Thermostatic PPO: +146.9%
- HDRL: +149.6%

Relative energy versus thermostatic PPO:

- HDRL annual difference: +1.1%

Why HDRL Is Still Worse than the Thermostatic Baseline

- Different objective:
 - thermostatic PPO is comfort-first and tracks a fixed target of 22 °C,
 - HDRL optimizes comfort + energy over the broader band [21,25] °C.
- Different information budget:
 - thermostatic PPO uses 17 features including forecasts and history,
 - HDRL uses only 5 current-state features.
- Different control logic:
 - HDRL relies on a hard seasonal switch at 12 °C,
 - this is too crude for shoulder seasons and mixed thermal regimes.
- Result:
 - HDRL may be acceptable as a structured controller,
 - but it is not yet better than the comfort-only baseline.

MORL Design in the Current Repository

The MORL entry point is `main.py`.

Current scalarization settings in `configs/env.yaml`:

- $w_{\text{comfort}} = 0.8$,
- $w_{\text{energy}} = 0.2$,
- energy scale = 2×10^{-4} ,
- comfort band = $[21, 25]^\circ\text{C}$.

Important current status:

- MORL is implemented,
- but by default it still trains directly on the BOPTEST backend,
- and it has not yet been fully re-evaluated after the direct- T_{Sup} redesign.

The repository also contains a surrogate-based safety filter in `layers/safety/action_filter.py`.

Logic:

- take PPO action,
- rollout the Neural ODE surrogate for a short horizon,
- check whether predicted temperatures stay inside a shrunk safe band,
- if unsafe, replace the action with a fallback controller.

In the current code:

- the horizon is typically 2–4 steps,
- a safety margin is used to compensate surrogate uncertainty,
- evaluation script: `evaluation/eval_safe_morl.py`.

Honest Status of MORL and Safe MORL

- The MORL stack and the safety layer are already implemented in code.
- However, after the direct- T_{Sup} redesign, the main validated results are still:
 - standard PI baseline,
 - thermostatic PPO baseline,
 - HDRL.
- Therefore, for this defense the MORL part should be presented as:
 - the next-stage objective,
 - architecturally ready,
 - not yet the final quantitative headline of the project.
- This is scientifically stronger than overclaiming incomplete results.

Main Contributions Across All Phases

- A full research pipeline from slow BOPTTEST RL to fast surrogate-assisted control.
- A physics-informed RC Neural ODE surrogate with large speedup and strong predictive accuracy.
- An inverse-calibration stage that materially improved physical fidelity.
- A structured Phase 3 experimental program that identified weather mismatch, seasonal mismatch, and action-response mismatch.
- A current direct- T_{Sup} redesign that produced a strong comfort baseline and a more honest benchmark structure.
- A codebase that already contains the next deployment layers: HDRL, MORL, and surrogate-based safety filtering.

- The current HDRL controller does not yet beat the thermostatic PPO baseline on the annual comfort-energy trade-off.
- The PI baseline remains far more energy-efficient because it solves a softer scheduled-comfort problem.
- MORL has not yet been fully re-trained and re-evaluated on the direct- T_{Sup} pipeline.
- The current HDRL gate is threshold-based rather than learned.
- The current annual benchmark uses 12 representative windows of 336 hours each rather than a full continuous-year rollout.

- Retrain MORL first on the direct- T_{Sup} surrogate, then fine-tune on BOPTEST.
- Replace hard seasonal switching with a learned gate or hysteresis-based meta-controller.
- Give HDRL and MORL the same richer temporal context as the thermostatic PPO baseline.
- Move from flat band-based comfort toward a target-aware comfort metric around 22 °C.
- Extend evaluation toward PMV/PPD and ASHRAE 55 style comfort metrics.
- Prepare edge deployment and real-building sensor integration after final controller stabilization.

Final Takeaways

- Phase 0 established the MORL baseline and exposed representation problems.
- Phase 1 delivered the key computational breakthrough: a fast and accurate Neural ODE surrogate.
- Phase 2 improved physical fidelity through inverse calibration.
- Phase 3 established that hierarchical structure, weather realism, and emergency logic matter, but also revealed a structural action mismatch.
- The current continuation solved that mismatch by moving to direct supply-air control.
- Today, the strongest validated result is the direct- T_{Sup} thermostatic PPO baseline, while HDRL is an operational but not yet dominant trade-off controller.
- The repository is now technically ready for the final MORL step.

Reproducibility: Key Files

- Report basis: docs/_tmp_morl_project_final_v3.txt extracted from MorlProjectFinalv3.pdf
- Surrogate model: surrogate/rc_node_v2.py
- Surrogate training: surrogate/train_surrogate_v2.py
- Direct- T_{Sup} data collection: data/collect_tsupply_data.py
- Thermostatic baseline: training/train_thermostatic.py
- HDRL training: training/train_hdrl.py
- Standard PI evaluation: evaluation/standard_controller_baseline.py
- Safety evaluation: evaluation/eval_safe_morl.py
- Current validated outputs:
 - outputs/thermostatic_yearly_summary.csv
 - outputs/hdrl_yearly_summary.csv
 - outputs/standard_controller_yearly_summary.csv
 - outputs/standard_controller_energy_comparison.csv

Reproducibility: Main Commands

- Thermostatic PPO:
 - `python training/train_thermostatic.py`
 - `PYTHONPATH=/app python3 evaluation/eval_thermostatic.py`
- HDRL:
 - `python training/train_hdrl.py`
 - `PYTHONPATH=/app python3 evaluation/yearly_validation_hdrl.py`
- Standard PI baseline:
 - `PYTHONPATH=/app python3 evaluation/standard_controller_baseline.py`
- MORL and safe MORL:
 - `SEED=42 python main.py`
 - `MODE=eval SEED=42 python main.py`
 - `PYTHONPATH=/app python3 evaluation/eval_safe_morl.py -model ...`

Thank you

Questions?